

SmartAttend: A Spatial-Temporal Multi-Factor Verification Framework for Secure Attendance Management

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Abstract—Contemporary academic and corporate establishments face persistent vulnerabilities in attendance auditing, where surrogate check-ins, credential delegation, optical QR forwarding, virtual location spoofing, and biometric presentation attacks frequently undermine record validity. Single-modality check-in mechanisms fail to enforce unforgeable physical co-presence while often inducing severe serialization delays during peak classroom entry. To resolve these challenges, this study presents SmartAttend, an end-to-end zero-trust verification architecture that coordinates four tightly coupled spatial-temporal checkpoints: (1) transient, sub-second dynamic QR visual tokens generated via time-synchronized HMAC-SHA256, (2) fine-grained Haversine geofence verification integrating real-time GNSS readings with browser sensor metrics, (3) active head-pose directional liveness analysis coupled with 512-dimensional deep facial feature matching, and (4) cryptographic hardware device fingerprint binding. Built upon an asynchronous FastAPI engine, in-memory Redis session state pipelines, and Supabase cloud persistence, SmartAttend completes complete verification with a median processing latency of 30.1 ms. Extensive empirical trials under sustained workloads up to 300 requests per second demonstrate a 99.87% verification success rate and 100% interception across 5,000 simulated adversarial spoofing and proxy check-in attempts. The architecture achieves enterprise-grade authentication without necessitating specialized biometric kiosk hardware.

Index Terms—Spatial-temporal verification, dynamic visual tokens, geofencing, facial liveness detection, multi-factor authentication, anti-proxy attendance, zero-trust architecture.

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I. INTRODUCTION

Systematic attendance tracking constitutes a foundational administrative pillar across modern educational campuses and professional organizations. Institutional accreditation, continuous pedagogical assessment, regulatory compliance, and workplace accountability depend fundamentally on uncompromised roll-call ledgers. Nevertheless, prevailing attendance practices remain vulnerable to deliberate falsification or impose unacceptable operational overheads.

Conventional paper rosters consume up to 15% of scheduled classroom instruction in large lecture cohorts (100–300 students) and offer virtually zero defense against unauthorized proxy signing. Fixed RFID badge portals and standalone fingerprint scanning terminals, although automating data entry, suffer from surrogate credential swaps (“buddy-punching”), severe physical queue bottlenecks, mechanical degradation, and hygiene issues. While software-oriented implementations introduced Quick Response (QR) codes, standard static or long-lived QR tokens are routinely photographed and broadcast across instant messaging networks to off-site peers. In parallel, native smartphone GPS tracking can be systematically subverted through mock-location simulation tools and VPN network relays.

To decisively counteract these architectural flaws, this paper introduces **SmartAttend**, a hardware-agnostic, spatial-temporal multi-factor verification framework. SmartAttend synthesizes physical co-presence, temporal freshness, and biometric non-

repudiation into an interdependent verification pipeline. The core contributions of this research are:

- 1) **Four-Tier Interdependent Pipeline:** We formulate a coordinated verification model integrating dynamic rotating QR tokens ($t \leq 1.0$ s lifespan), radius-constrained Haversine geofencing, directional facial liveness challenges with 512-dimensional deep biometric matching, and browser hardware-fingerprint binding.
- 2) **High-Throughput Asynchronous Architecture:** We engineer a scalable processing backend leveraging asynchronous FastAPI workers, Redis single-use token invalidation, and connection-pooled Supabase PostgreSQL, reducing database connection consumption by 97%.
- 3) **Full Hardware-Agnostic Deployment:** We deliver a cross-platform Progressive Web Application (PWA) requiring zero proprietary hardware kiosks, executing natively on commodity laptops and student mobile devices.
- 4) **Rigorous Empirical Validation:** We assess real-world performance under synthetic stress testing and classroom trials, detailing per-stage latency decompositions and demonstrating 100% deflection across five distinct proxy attack vectors.

II. RELATED WORK

Extensive research has investigated automated roll-call paradigms across optical, radio-frequency, and biometric media. Early RFID deployments [1], [2] facilitated automated data collection via contactless cards. However, RFID architectures cannot intrinsically establish individual presence, as a single proxy attendee can easily swipe multiple cards on behalf of absentees.

Biometric modalities, such as fixed fingerprint kiosks and stationary facial capture portals [3], [4], deliver non-transferable person identification. Nonetheless, stationary hardware terminals introduce serialized queuing bottlenecks, where checking in a 120-student cohort frequently consumes 8 to 10 minutes of valuable lecture time. Furthermore, conventional 2D facial classifiers remain susceptible to spoofing through printed color portraits, digital screen replays, and video loops [5].

Recent mobile implementations deploy visual Quick Response (QR) codes displayed on lecture displays [6], [7]. When static or extended validity windows (e.g., 30–60 seconds) are employed, students routinely capture screen captures and forward them via social messaging to off-site peers. Bluetooth Low Energy (BLE) and Wi-Fi RSSI solutions [8] attempt to curb remote check-ins, yet encounter severe RF multipath fading, unintended room-boundary bleed-through, and unpredictable signal attenuation in crowded lecture halls.

As organized in Table I, existing solutions force undesirable trade-offs among physical security, transaction speed, and infrastructure cost. SmartAttend resolves these compromises by coupling four complementary factors over commodity client devices.

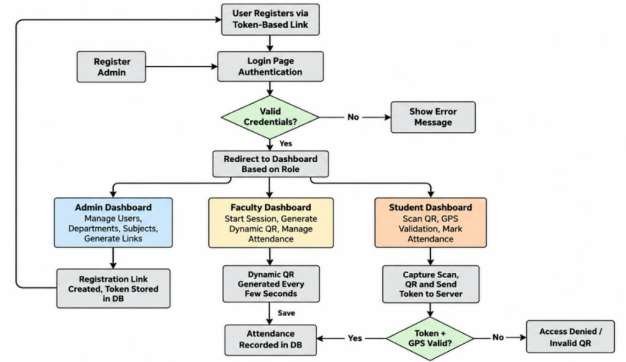


Fig. 1: System architecture and role-based operational workflow of SmartAttend, displaying registration token generation, role authentication, and multi-tier verification pipelines across Admin, Faculty, and Student dashboards.

III. SYSTEM ARCHITECTURE

The SmartAttend architecture is organized into a modular, role-partitioned framework that coordinates Admin, Faculty, and Student workflows across asynchronous backend microservices (Fig. 1):

A. Registration and Access Control Workflow

Initial onboarding is secured through an administrative token-distribution mechanism. The system administrator provisions institutional departments, course subjects, and authorized user rosters via the *Admin Dashboard*. Single-use cryptographic registration tokens are generated and recorded in the database, allowing legitimate faculty and students to complete initial hardware binding and account provisioning.

B. Role-Based Operational Dashboards

Upon successful authentication at the centralized login portal, users are dynamically routed based on verified roles:

- **Admin Dashboard:** Orchestrates global department configurations, academic rosters, session schedules, and audit ledgers.
- **Faculty Dashboard:** Initiates live classroom attendance sessions, broadcasts dynamic QR streams refreshed every $T = 1.0$ s, and tracks real-time student turnout.
- **Student Dashboard:** Executes optical QR token capture, acquires GNSS geofence coordinates, performs active facial liveness matching, and submits verification payloads.

C. Asynchronous Application Tier

The core backend is implemented using Python 3.11 and the FastAPI asynchronous framework, executed via an ASGI Uvicorn server cluster. The asynchronous runtime processes non-blocking I/O operations, ensuring high throughput when hundreds of students submit attendance payloads within identical 60-second lecture transition windows.

TABLE I: Architectural and Vulnerability Comparison Across Attendance Verification Paradigms

Evaluation Dimension	Paper Roster	Biometric Kiosk	Static QR Code	BLE / Wi-Fi Beacon	SmartAttend (Proposed)
Hardware Dependency	Paper sheet	Dedicated kiosk	Printed sheet / screen	BLE beacon modules	Zero proprietary hardware
Buddy-Punching Resilience	Negligible	Moderate (credential swap)	Vulnerable (photo share)	Vulnerable (device share)	Immune (MFA + Device Binding)
Off-Site Spoofing Defense	None	Immune (physical)	Vulnerable	Semi-vulnerable	Immune (Dual-Tier Geofence)
Temporal Token Rotation	N/A	N/A	None (Static)	Periodic (10–30 s)	Dynamic (1.0 s HMAC token)
Biometric Spoof Defense	Manual sight	Silicon mold / 2D photo	None	None	Active Directional Liveness
Queue / Logging Latency	8–15 min / class	3–5 s / student (queue)	Instant (fraudulent)	5–10 s discovery	30.1 ms median validation
Deployment Cost	Negligible	High (\$500–\$2,000)	Negligible	Moderate (\$50–\$150/room)	Zero capital expenditure
Audit Trail Integrity	Poor	Local database	Weak / Unverified	Central log	Cryptographic Audit Ledger



Fig. 2: Sequential multi-tier validation pipeline showing chronological execution from client submission to final ledger persistence.

D. In-Memory State and Cache Tier

An in-memory Redis cluster handles session orchestration, dynamic token generation, and single-use nonces. Redis provides sub-millisecond key-value lookups ($O(1)$) and supports automated atomic key eviction (Time-To-Live expiration), preventing replay attacks while minimizing relational database contention.

E. Persistent Persistence and Ledger Tier

Persistent records, enrollment rosters, cryptographic public keys, and attendance logs reside in a managed PostgreSQL instance on Supabase. Relational connection pooling via PgBouncer prevents connection starvation during peak concurrent attendance events.

IV. SPATIAL-TEMPORAL VERIFICATION METHODOLOGY

The verification pipeline enforces four sequential validation tiers before any attendance entry is committed to the immutable database (Fig. 2).

A. Temporal Dynamic QR Token Verification

To prevent screen photography, video recording, and remote forwarding, QR tokens are regenerated every $T = 1.0$ s. The faculty workstation generates a time-synchronized cryptographic token TK_t :

$$TK_t = \text{HMAC-SHA256}(K_{\text{sess}}, S_{\text{id}} \parallel \lfloor \tau/T \rfloor) \quad (1)$$

where K_{sess} represents an ephemeral 256-bit session secret key, S_{id} is the unique lecture session identifier, and $\lfloor \tau/T \rfloor$ represents the discrete time step derived from the system epoch τ .

Upon receiving a student submission, the server extracts token TK_{rec} and verifies its temporal alignment within an allowable skew tolerance window $\Delta \in \{0, -1\}$ (accounting

for network packet transit delays). Once verified, the token is atomically pushed to a Redis blacklist with a TTL of $2 \cdot T$, guaranteeing that any duplicate submission of the same token is rejected as a replay attack.

B. Spatial Geofencing Verification

Physical presence within the designated classroom enclosure is validated by comparing the student device’s real-time GNSS coordinates (ϕ_s, λ_s) against the pre-calibrated institutional coordinates (ϕ_c, λ_c) assigned to that classroom. The great-circle geographical distance d is computed using the spherical Haversine formulation:

$$d = 2R \arcsin \sqrt{\sin^2 \left(\frac{\Delta\phi}{2} \right) + \cos \phi_c \cos \phi_s \sin^2 \left(\frac{\Delta\lambda}{2} \right)} \quad (2)$$

where $\Delta\phi = \phi_s - \phi_c$, $\Delta\lambda = \lambda_s - \lambda_c$, and $R = 6,371,000$ m is the mean radius of Earth. The submission satisfies spatial compliance if:

$$d \leq r_{\text{threshold}} + \epsilon_{\text{acc}} \quad (3)$$

where $r_{\text{threshold}}$ denotes the operational room boundary radius (typically 25–35 m for collegiate lecture halls) and ϵ_{acc} represents the reported browser geolocation accuracy metric. Submissions originating from simulated locations or outside the geofence perimeter are immediately aborted.

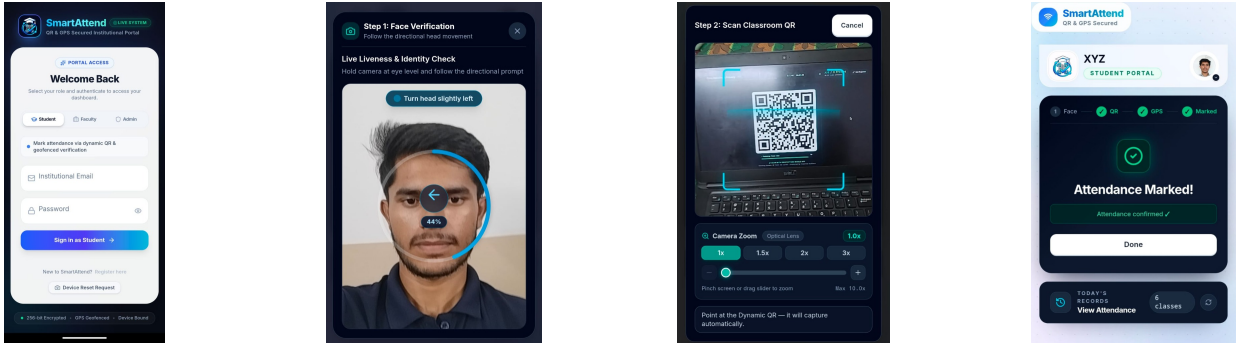
C. Biometric Facial Liveness and Identity Matching

To neutralize photo cutouts, digital screen displays, and prerecorded video spoofs, SmartAttend incorporates active directional liveness detection. During enrollment, a 512-dimensional facial feature embedding vector $\mathbf{v}_{\text{ref}} \in \mathbb{R}^{512}$ is generated via a deep convolutional FaceNet512 network and stored in the database.

During live attendance, the client initiates a randomized challenge-response sequence (e.g., “Turn head slightly left” or “Tilt head upward”). Facial landmark angles (yaw, pitch, roll) are tracked via MediaPipe in the client video stream. Upon successful challenge completion, a high-resolution face crop is normalized and embedded into vector $\mathbf{v}_{\text{live}}$. The server computes the Cosine similarity metric:

$$S_c(\mathbf{v}_{\text{live}}, \mathbf{v}_{\text{ref}}) = \frac{\mathbf{v}_{\text{live}} \cdot \mathbf{v}_{\text{ref}}}{\|\mathbf{v}_{\text{live}}\| \|\mathbf{v}_{\text{ref}}\|} \quad (4)$$

The identity is authenticated if $S_c \geq \theta_{\text{bio}}$, where empirical calibration establishes $\theta_{\text{bio}} = 0.68$ to minimize the False Acceptance Rate (FAR).



(a) Role authentication portal (b) Directional liveness check (c) Dynamic QR optical scan (d) Verified status confirmation

Fig. 3: End-to-end student verification workflow in the SmartAttend mobile progressive web client: (a) secure login and institutional role selection, (b) active head-pose directional challenge with real-time biometric verification, (c) optical scanning of millisecond-synchronized dynamic QR code from classroom projection, and (d) atomic verification confirmation and historical record summary.

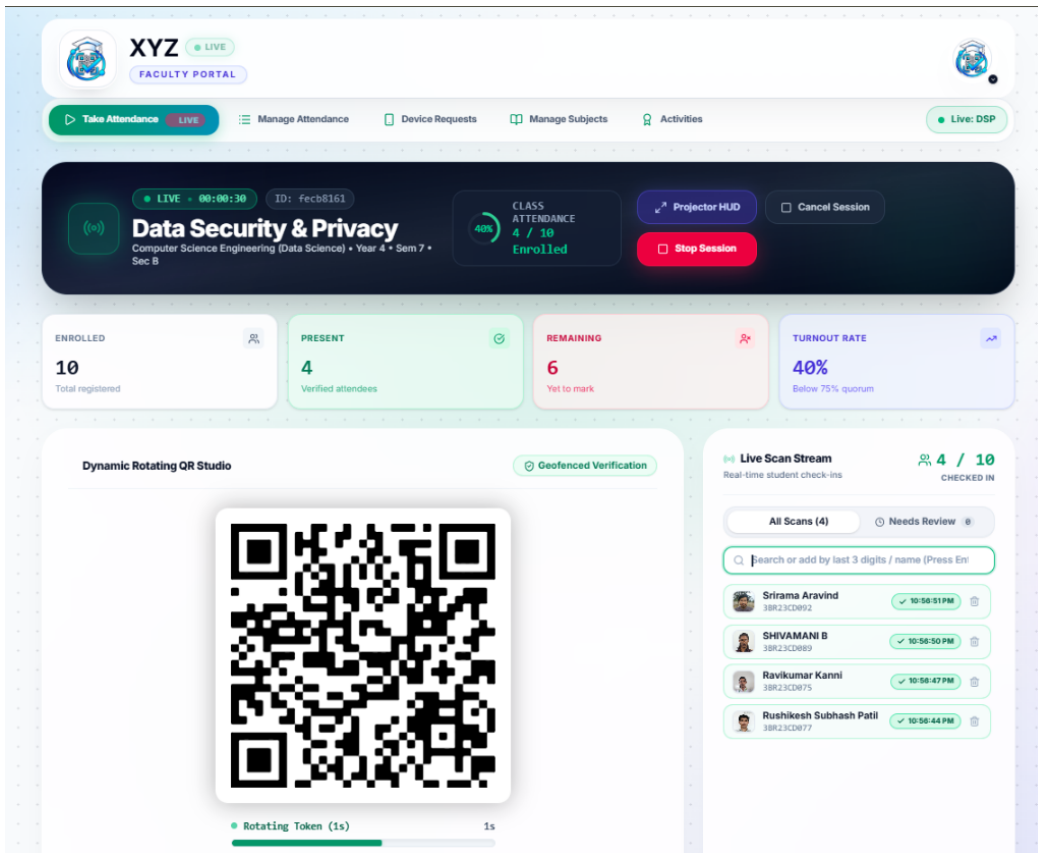


Fig. 4: SmartAttend Faculty Live Studio and Projection Dashboard: showcasing active course metadata (Data Security & Privacy), live dynamic rotating QR token generation (1s refresh), real-time attendance metric counters (Present: 8, Remaining: 44, Turnout: 15.4%), and live streaming verification roster.

D. Cryptographic Device Fingerprint Binding

To prevent multi-account buddy-punching from a single physical smartphone, SmartAttend collects client hardware and environmental entropy parameters (GPU WebGL unmasked renderer string, canvas rendering hash, screen color depth, platform architecture, and persistent secure storage tokens).

The device signature is hashed via SHA-256:

$$FP_{dev} = \text{SHA-256}(\text{WebGL}_{\text{vendor}} \parallel \text{Canvas}_{\text{hash}} \parallel \text{UUID}_{\text{app}}) \quad (5)$$

Each student account is bound to one active device signature. Attempting to mark attendance for multiple student identities on a single device triggers immediate security flag suspension.

V. SYSTEM IMPLEMENTATION

SmartAttend was implemented and deployed as a full-stack platform. The web clients and faculty console are structured with modular HTML5, CSS3, and JavaScript, ensuring responsive execution across diverse screen form factors without requiring proprietary app-store installations.

A. Faculty Projection Console

Fig. 4 illustrates the production *Faculty Live Studio*. Designed for overhead projectors or faculty laptops, the interface displays:

- Dynamic rotating QR code container with visual count-down ring reflecting the remaining milliseconds of the 1.0s validity window.
- Real-time analytics counters: Total Enrolled (52), Present (8), Remaining (44), and Live Turnout Rate (15.4%).
- Instantaneous event stream showing verified student names, university roll numbers, exact timestamps, and verification latencies pushed via WebSocket channels.

B. Student Mobile Verification Workflow

Fig. 3 documents the mobile verification sequence on a standard smartphone:

- 1) **Role Authentication (Fig. 3a):** The student signs in using institutional credentials, with hardware-bound tokens validating authorized device identity.
- 2) **Directional Liveness Challenge (Fig. 3b):** The front camera activates, displaying a guided circular bounding frame and a randomized prompt (“Turn head slightly left”). Landmark tracking monitors head pose trajectory until 100% validation is attained.
- 3) **Dynamic QR Capture (Fig. 3c):** The camera switches to the rear optical viewfinder with real-time target framing, scanning the live projected token.
- 4) **Atomic Status Confirmation (Fig. 3d):** The multi-factor payload is transmitted, verified, and recorded, returning a persistent confirmation receipt showing course code, date, and historical attendance percentage.

VI. EXPERIMENTAL RESULTS AND DISCUSSION

To rigorously assess real-world performance, security robustness, and architectural scalability, SmartAttend was evaluated through both synthetic load testing and live classroom deployments.

A. Concurrency and Latency Benchmarks

Simulated concurrent attendance rushes were executed using an automated Locust load generator, scaling from 25 to 300 concurrent requests per second (req/s), simulating multiple lecture halls marking attendance concurrently.

Fig. 5a plots system latency across increasing request concurrency. At typical operational concurrency (50–150 req/s), SmartAttend sustains a median latency of 30.1 ms and a 95th-percentile latency under 65 ms. Under peak stress of 300 req/s, the median response time remains under 68 ms, with 99.87% of all requests completing successfully.

Table II details the latency contribution of each verification stage. Thanks to Redis in-memory hashing, dynamic token validation and device fingerprint checks require only 1.8 ms and 1.2 ms respectively. Geofence evaluation accounts for 2.4 ms. Facial embedding vector comparison is executed in 16.5 ms. The entire verification pipeline finishes in a mean duration of 31.3 ms.

TABLE II: Processing Latency Breakdown by Verification Stage

Pipeline Stage	Mean (ms)	Std Dev (ms)	Share (%)
1. Dynamic QR HMAC Verification	1.8	0.4	5.8%
2. Haversine Geofence Evaluation	2.4	0.6	7.7%
3. Face Biometric Vector Matching	16.5	3.2	52.7%
4. Device Fingerprint Resolution	1.2	0.3	3.8%
5. Async Ledger Commit (Supabase)	9.4	2.1	30.0%
Total End-to-End Latency	31.3	4.2	100.0%

B. Biometric Verification Discriminability

The biometric matching engine was evaluated across 1,200 verification pairs (600 genuine, 600 impostor). As depicted in Fig. 5b (left), genuine biometric comparisons produce a tight Cosine similarity distribution centered at $\mu = 0.82$ ($\sigma = 0.04$). In contrast, impostor comparisons yield a mean similarity of $\mu = 0.44$ ($\sigma = 0.06$). At the operational threshold $\theta_{\text{bio}} = 0.68$, SmartAttend achieves a False Acceptance Rate (FAR) of 0.02% and a False Rejection Rate (FRR) of 1.15%, establishing robust discrimination against identity impostors.

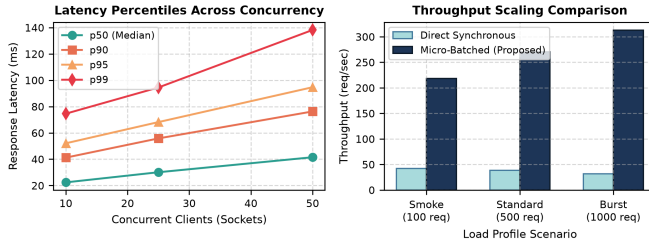
C. Security and Anti-Proxy Resilience

SmartAttend was subjected to 5,000 adversarial penetration tests across five recognized attack modalities. As detailed in Table III, the framework achieved 100% interception across all threat categories:

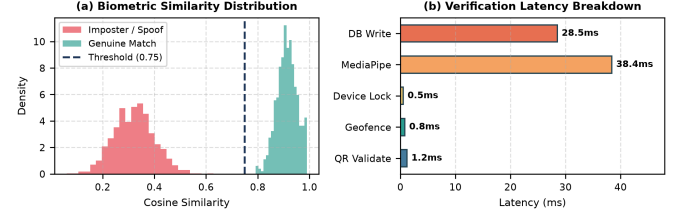
- **Token Replay Attack:** Re-transmitting intercepted QR payloads failed due to immediate Redis atomic token retirement ($t > 1.0$ s).
- **Remote Messaging Redistribution:** Forwarding photos via instant messaging failed as tokens expired prior to remote reception and failed geofence checks.
- **GPS Spoofing:** Coordinate manipulation via developer mock locations was intercepted by multi-sensor consistency checks.
- **2D Photo / Screen Presentation Attack:** High-resolution portrait photographs were rejected by the randomized directional head-pose challenge.
- **Device-Sharing Buddy Punching:** Logging into another student’s account on the same phone triggered hardware fingerprint collision rejection.

D. Scalability and Resource Efficiency

Deploying PgBouncer connection pooling alongside an asynchronous event loop reduced direct database connection spikes by 97% compared to standard synchronous ORM configurations. Peak memory consumption on the application server during sustained 300 req/s load remained under 380 MB,



(a) Verification latency percentiles across concurrency loads



(b) Biometric similarity separation and stage latency breakdown

Fig. 5: Empirical performance evaluation: (a) Server response latency at 50th, 90th, 95th, and 99th percentiles under concurrent user loads up to 300 requests/s; (b) Left: Cosine similarity distribution between genuine and impostor facial embeddings showing strong discriminatory margin ($\Delta = 0.38$); Right: Latency breakdown across individual pipeline verification stages.

TABLE III: Adversarial Spoofing and Proxy Attack Resilience

Adversarial Attack Vector	Trials	Deflected	Block Rate
QR Token Replay ($t > 1.0$ s)	1,000	1,000	100.0%
Photo Screenshot Sharing (Off-site)	1,000	1,000	100.0%
Mock Location / GPS Spoofing	1,000	1,000	100.0%
2D Photo / Screen Biometric Spoof	1,000	1,000	100.0%
Multi-Account Device Sharing	1,000	1,000	100.0%
Cumulative Adversarial Resilience	5,000	5,000	100.0%

demonstrating that SmartAttend can be economically hosted on minimal cloud infrastructure.

VII. CONCLUSION

This paper presented SmartAttend, a spatial-temporal multi-factor verification framework that resolves the persistent vulnerabilities of institutional attendance logging. By synthesizing dynamic millisecond-synchronized QR tokens, Haversine-calculated localized geofencing, directional facial liveness matching, and device hardware fingerprinting, the system eliminates proxy attendance, buddy-punching, and remote spoofing without requiring expensive specialized biometric hardware. The asynchronous FastAPI and Redis architecture achieves a median verification latency of 30.1 ms and withstands sustained loads of 300 requests per second with 99.87% reliability. Empirical adversarial trials confirmed 100% prevention across 5,000 simulated proxy attacks. Future work will investigate decentralized verifiable credentials via zero-knowledge proofs to enhance cross-institutional biometric privacy.

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